Nutrition Atlas

Carbohydrates

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Carbohydrates are one of the three main macronutrients required for the proper body functioning, along with proteins and fats. These are in form of sugar or starches and are the key fuel to our body. Also called "quick energy" foods as they are easily broken down into glucose, which can either be used right away or gets converted to glycogen and stored in the liver and muscles for later use.

Carbohydrates dominate diets across the world in the form of staples as rice, wheat, grains, roots and tubers.

Classification of carbohydrates

Based on the digestion, carbohydrates can be classified as:

Simple Carbohydrates

Out of the total calories consumed, 50-55% should come from carbohydrate.

The quantity and type of carbohydrates included in the diet are important to maintain good health.

Each gram of CHO gives 4 Kcal.

Simple carbohydrates when ingested are quickly digested and absorbed thereby raising the blood glucose levels rapidly.

Natural sugars are present in foods like fruits, vegetables, and milk. They are healthy due to the presence of vitamins, minerals and fibre complexed with the natural sugars.

Unlike natural sugars, processed foods like candies, pastries, biscuits, and sugary beverages are high in added refined sugars. They pose some serious health concerns like weight gain, diabetes, heart problems, etc.

Foods such as white bread, white rice and flour are made after a lot of processing of whole grains. This leads to loss of nutrients especially fibre leading to spike in blood sugar.

Complex Carbohydrates

Starches are complex carbohydrates which take longer to get digested and absorbed thereby releasing glucose slowly into the blood stream.

Foods that are high in starch includes whole grains, legumes and starchy vegetables.

The unprocessed or minimally processed whole grains, vegetables, fibre-rich fruits and beans are good sources of healthy carbohydrates.

Fibre, a type of complex carbohydrate is present in fruits, vegetables, whole grains, nuts and legumes.

What are the functions of carbohydrates?

It is the major source of energy in the diet, and under normal condition it is the sole source of energy for the vital functions of key organs especially brain.

It is stored form of energy in liver and muscles as glycogen. Glycogen breaks down to produce glucose to provide constant supply of energy.

Protein sparing action: Sufficient intake of carbohydrates prevent breakdown of protein for required energy and retain them for body building.

It is necessary for normal fat metabolism. When the carbohydrate intake is low, body excessively uses fat for energy, that can produce ketone bodies.

Adequate intake of carbohydrates is necessary to maintain vital bodily functions of heart and central nervous system.

How much carbohydrates should be included in a healthy diet?

Note: In lens of rising prevalence of overnutrition and obesity, it is to be noted that excess carbohydrate intake will be stored as fat in body!

In general carbohydrates should contribute to about 45-65 % of the daily caloric intake, which roughly equals to 200 – 300 g of carbohydrates per day.

ICMR-NIN recommends a minimum intake of 100 – 130 g of carbohydrates per day for ages 1 year and above.

Which kind of carbohydrates are good for health?

Complex carbohydrates, found in unprocessed or minimally processed cereals & grains, fruits, and vegetables, are beneficial, whereas simple carbohydrates in highly processed foods, refined flours, sugary drinks, and sweets should be avoided due to their tendency to cause rapid spikes in blood glucose levels.

What are the signs and symptoms of carbohydrates deficiency?

Deficiency of carbohydrates occurs as part of overall energy undernutrition.

Carbohydrate deficiency can be manifested in the body as depletion of body fat, loss of muscle mass, loss of protein from heart, liver and kidney, impaired immune response, infections and reduced absorption of available nutrients.

What happens when carbohydrates intake is low?

Hypoglycemia – can occur due to fasting, alcohol intake, inadequate energy intake, poor food habits.

Lactose intolerance* - due to absence of enzyme lactase that converts milk sugar and lactose to glucose and galactose.

Note: Proper diagnosis by trained medical professional is crucial to identify intolerance

What happens when there is overconsumption of carbohydrates?

Excessive carbohydrate consumption, especially of refined carbohydrates like sugary foods and drinks, can increase the risk of developing obesity and chronic diseases such as type 2 diabetes, cardiovascular disease, and metabolic syndrome.

Nutrition label

On the nutrition label, the term "total carbohydrate" refers to all three forms of carbohydrates: starch, sugar, and fiber.

The term ‘total carbohydrate’ on the nutrition label includes all the three types of carbohydrates i.e., Starch, sugar and fiber.

Total Sugars in the Nutrition Facts label indicates both the natural sugar present in food and added sugar in the product.

Added sugars indicates the sugars and syrups added during processing of the product. Terms such as fructose, dextrose, cane juice, corn syrup, maple syrup and honey are also considered added sugars.

GI and GL of foods

Glycaemic index (GI)

Glycaemic index (GI) is a numerical value given to impact of carbohydrate-containing foods on blood glucose levels in comparison to another carbohydrate food (usually 50gm of glucose).

The slower the rise in blood glucose after eating a food, the lower is its glycaemic index and vice versa.

The GI ranges from 0 to 100.

In general, a food's GI increases with processing, while a food's GI decreases with fiber or fat content.

GI category	GI value	Examples / description
Low	55 or lower	Whole foods such as unrefined grains, non-starchy vegetables, and fruits tend to have a lower GI.
Medium	56–69	Dosa, idli, Pongal, Muesli, and Millet porridge.
High	70 or above	Processed foods made with refined sugar and flour such as candy, bread, cake, and cookies. White Rice is also high in GI.

Glycaemic load (GL)

Glycaemic load (GL) is a measure of the impact of food on the blood sugar levels based on GI of the food and the amount of carbohydrates consumed.

It is a more reliable indicator than GI.

It is calculated as GI/100× CHO content of the food.

A high GI food if consumed in small amounts will give lower glycaemic load and vice-versa.

A diet with low GI and GL can help in controlling blood sugar levels in type 2 diabetes individuals.

GL category	GL value	Examples
Low GL	10 or lower	Wheat roti, pulses and legumes, most fruits and vegetables.
Medium GL	11–19	Brown rice, Dhokla, boiled potato, sweet potato, banana, papaya.
High GL	20 or higher	Poori, White rice, Idli, Dosa, French fries.

Rich sources of Carbohydrates

- Cereals and millets
- Pulses and legumes
- Oils, Fats, Nuts & oilseeds
- Vegetables
- Roots and tubers
- Green leafy vegetables
- Fruits
- Spices and Herbs
- Dairy

When planning your meals

Choose foods that have a low to medium GI.

When eating a high GI, combine it with low GI foods to balance the effect on your glucose levels.

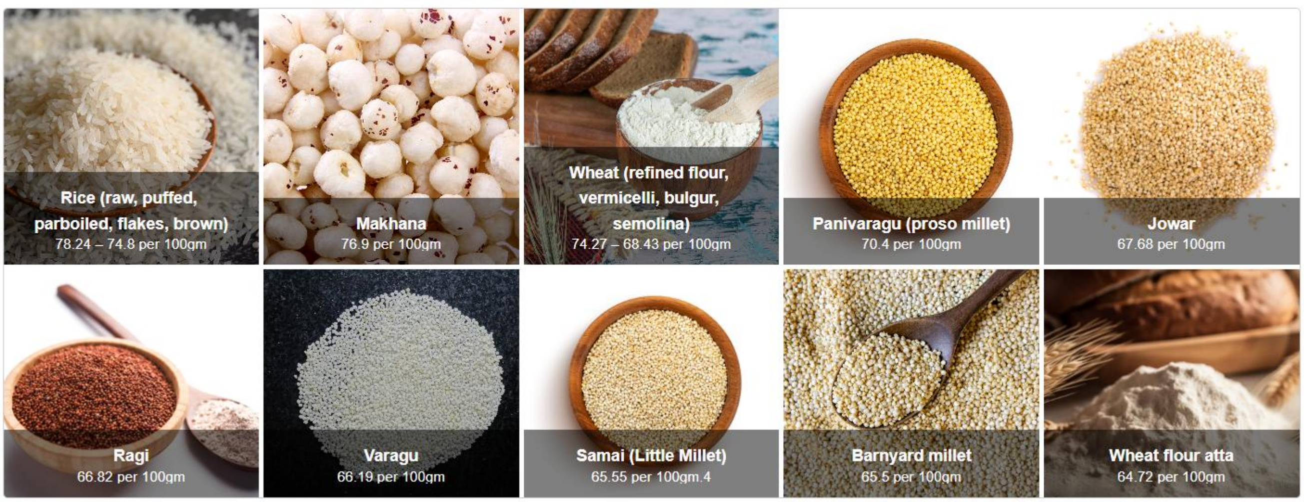
Portion size of the food consumed also determines the glycaemic response.

The glycaemic response of the food depends on a number of factors such as type of carbohydrate, amount of carbohydrate, amount of protein and fat, food matrix, nutrient interactions, type of food processing, other food components, etc. (RDA, 2020).

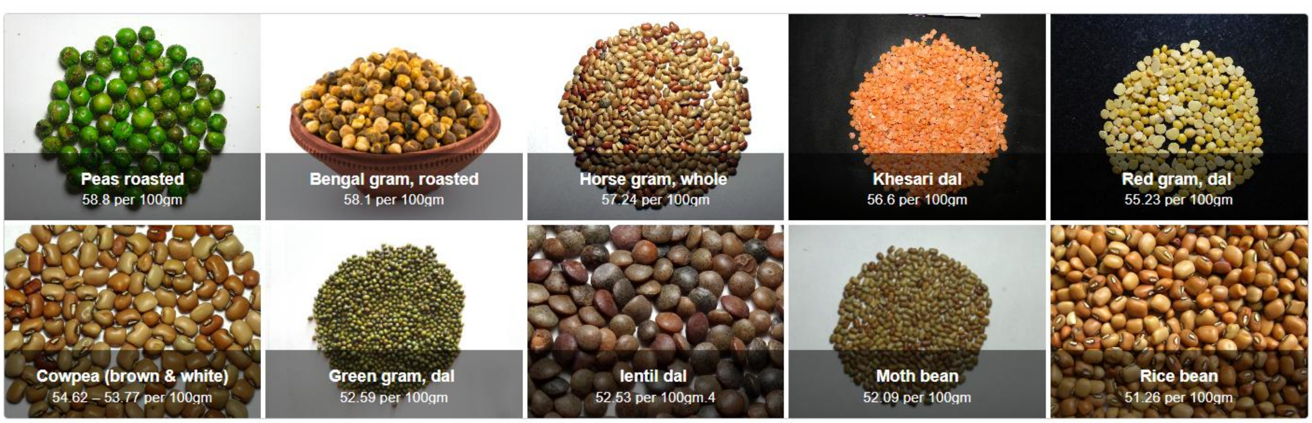
Rich sources of Carbohydrates

Top Sources

Cereals and millets



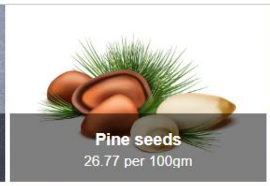
Pulses and legumes

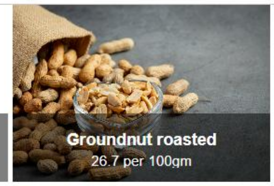


Oils, Fats, Nuts & oilseeds


Garden cress, seeds
33.66 per 100gm


Sunflower Seeds
30.18 per 100gm


Pine seeds
26.77 per 100gm


Groundnut roasted
26.7 per 100gm


Cashew nut
25.46 per 100gm


Niger seeds (black & gray)
22.98 – 20.59 per 100gm


Ground nut
17.27 per 100gm


Mustard seeds
16.8 per 100gm.4


Pistachio nuts
15.82 per 100gm


Pumpkin seeds
15.6 per 100gm

Vegetables


Water chestnut, dry
68.9 per 100gm


Karonda dry
67.1 per 100gm


Sundakai, dry
55 per 100gm

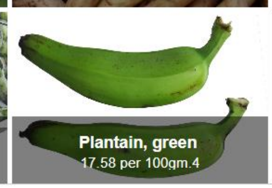

Lotus stem, dry
51.4 per 100gm


Sea weeds,dry
51.2 per 100gm


Beans, scarlet runner
28.9 per 100gm


Jackfruit seed
25.8 per 100gm


Red gram, tender, fresh
19.46 per 100gm


Plantain, green
17.58 per 100gm.4


Peas, fresh
11.88 per 100gm

Fruits


Dates (dark brown and pale brown)
84.87 per 100gm

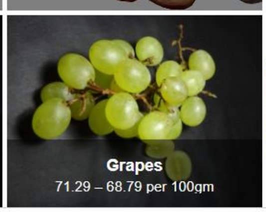

Apricot, dried
78.24 – 74.8 per 100gm

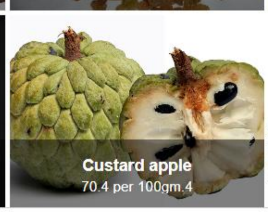

Raisins, dried
76.9 per 100gm


Dates, processed
74.27 – 68.43per 100gm


Bael fruit
74.91 – 72.67 per 100gm


Banana (red, montham, robusta, poovam)
72.63 per 100gm


Grapes
71.29 – 68.79 per 100gm


Custard apple
70.4 per 100gm.4


Rambutan
67.95 per 100gm


Fig
67.68 per 100gm

Spices and Herbs

Dairy

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